AWS Cross-Region Disaster Recovery Plan

Introduction

The objective of this presentation is to design a cost-effective cross-region disaster recovery (DR) solution for a payment system while minimizing data loss and ensuring high reliability. The solution incorporates AWS services to balance cost and performance, considering near-zero Recovery Point Objective (RPO) and acceptable Recovery Time Objective (RTO).

1. Cross-Region Disaster Recovery (DR) Solution

Proposed Architecture

- Primary Region:
  - Application Layer: Deployed in Amazon ECS (Fargate) or EKS for container orchestration.
  - Database Layer: Amazon RDS Multi-AZ for the primary database with read replicas.
  - Storage Layer: Amazon S3 with cross-region replication enabled.
  - Cache Layer: Amazon ElastiCache (Redis).
  - Monitoring: Amazon CloudWatch and AWS X-Ray for tracing.

- Secondary (DR) Region:
  - Cold Standby:
    - Database: Amazon RDS read replica, ready for promotion to primary.
    - Application Layer: AMIs and configuration stored in S3 or Systems Manager Parameter Store for quick deployment.
    - ElastiCache: Redis snapshot replicated for restoration.
    - DNS Failover: Amazon Route 53 with health checks for automatic redirection.
  - Data Synchronization:
    - Cross-region replication for S3 and DynamoDB tables.
    - Asynchronous replication for RDS read replicas.

Failover Workflow

1. Monitor the health of the primary region using Route 53 health checks.

2. Promote the secondary region's RDS read replica to primary and restore services.

3. Redirect traffic to the secondary region using Route 53.

2. Cost-Effective DR Approach

Cost Optimization Strategies

- Storage:
  - Enable lifecycle policies for S3 to move data to infrequent access (IA) or Glacier Deep Archive.
- Compute:
  - Use on-demand instances in the primary region and spot instances for less critical workloads in the DR region.
  - Keep ECS task definitions and infrastructure IaC scripts ready for DR activation instead of running resources actively in DR.
- Database:
  - Use read replicas in the DR region for RDS instead of active multi-region RDS deployments.
- Monitoring:
  - Use consolidated CloudWatch metrics and logs across regions.

3. Data Loss Considerations

- RPO (Recovery Point Objective): Near-zero data loss achieved via:
  - Cross-region replication for S3, DynamoDB (using global tables), and RDS.
  - Real-time database replication using AWS Database Migration Service (DMS) in Change Data Capture (CDC) mode.
- RTO (Recovery Time Objective): Target RTO of under 15 minutes with:
  - Pre-configured CloudFormation templates or Terraform scripts.

- Pre-defined Route 53 health checks for automated failover.

4. Assumptions

- The primary region handles 1TB of database data, with daily changes of 10%.
- 10 million monthly API requests with 80% reads and 20% writes.
- 100GB/month of static files stored in S3.
- Cross-region latency between regions is <100ms.

5. Cost Estimation

Primary Region

Service	Usage	Cost (Monthly)
Amazon RDS	1TB, db.m5.large	$230
Amazon S3 (Standard)	1TB	$23
S3 Data Transfer	100GB	$9
ECS (Fargate)	500 vCPU hours	$120
CloudWatch	100GB logs	$10
Total		$392

DR Region

Service	Usage	Cost (Monthly)
Amazon RDS (Replica)	1TB, db.m5.large	$115
Amazon S3	1TB	$12
Route 53 (Health)	10 checks	$5
DynamoDB	100GB	$15
Total		$147

Total Monthly Cost (Both Regions): ~$539.

Tools and Rationale

- Amazon S3: Reliable, cost-efficient storage with cross-region replication.
- Amazon RDS: Ensures ACID compliance and near-real-time replication.
- Amazon Route 53: Simple and cost-effective DNS failover.
- CloudFormation/Terraform: Simplifies infrastructure provisioning.
- AWS DMS: Facilitates near-real-time data replication.

Conclusion

This DR solution balances cost and performance using AWS's inherent scalability and resilience features. It minimizes downtime and data loss while ensuring seamless failover with an emphasis on cost-effective practices.